

Progress Test 3.B — Thermo-S1

3.B · The ideal gas and Clapeyron's law

Semester 1 · Academic Year 2026–2027

Family name: Given name:

Duration: 10 minutes. *No documents. Calculator not allowed.* Answer inside the printed frames only; the margin boxes are for the marker. **Total: 14 marks.**

Exercise 1. Units

(1 mark)

Express the unit of the gas constant R , $\text{J K}^{-1} \text{mol}^{-1}$, in SI base units.

[illegible]

Marking

1	
0	1

Exercise 2. The ideal gas — definition and Clapeyron's law

(5 marks)

Define the ideal gas model. Write Clapeyron's law, giving the SI unit of every quantity and the value of R .

[illegible]

Marking

①		
0	1	2

②		
0	1	2

③	
0	1

Exercise 3. Controlled and free variables

(3 marks)

For a hot-air balloon (opening at the bottom, rigid envelope), the variables held fixed are:

- ☐ n and V
☐ p and V
☐ n and T
☐ n and p

A weather balloon (sealed, flexible) rises; the outside pressure decreases while its temperature stays about the same. Its volume:

- ☐ increases
 ☐ decreases
 ☐ stays the same
 ☐ becomes zero

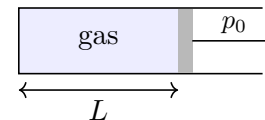
1.0 mol of gas at 300 K and 1.0×10^6 Pa occupies 2.5 L. With $RT \approx 2.5 \times 10^3 \text{ J mol}^{-1}$, its compressibility factor $Z = pV/(nRT)$ is about:

- ☐ 0.1
 ☐ 1.0
 ☐ 10
 ☐ 2.5

Exercise 4. Piston and cylinder

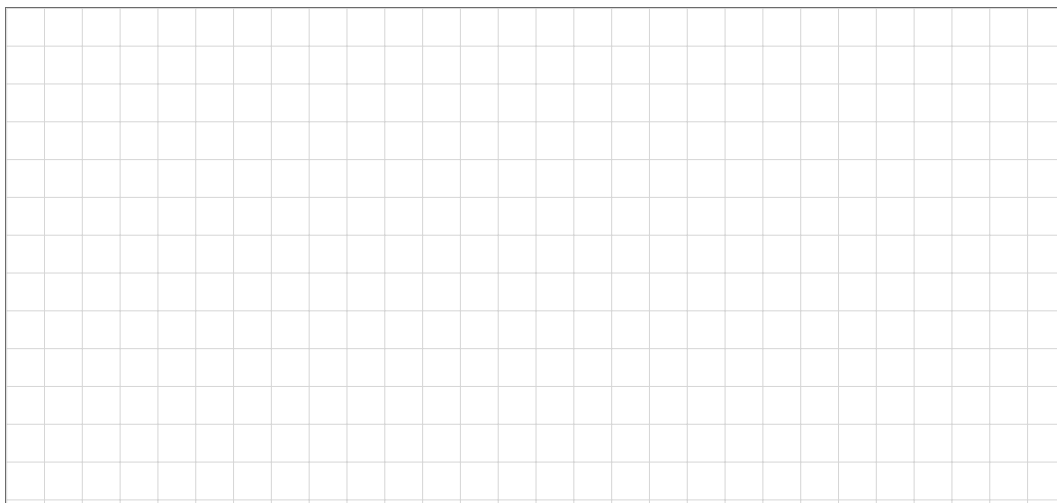
(5 marks)

A horizontal cylinder of cross-section S contains n moles of air, treated as an ideal gas. The piston is initially at L_1 and the temperature is T_1 .



(1) Compute p_1 . (2) The gas is compressed isothermally to L_2 : compute p_2 . (3) The piston is locked at L_2 and the gas is heated to T_3 : compute p_3 . (4) The piston is released against the atmosphere p_0 , at $T_4 = T_3$: compute V_4 and L_4 .

$S = 50 \text{ cm}^2$	$n = 0.10 \text{ mol}$	$L_1 = 50 \text{ cm}$	$L_2 = 25 \text{ cm}$	$T_1 = 300 \text{ K}$
$T_3 = 450 \text{ K}$	$p_0 = 1.0 \times 10^5 \text{ Pa}$	$RT \approx 2.5 \times 10^3 \text{ J mol}^{-1} \text{ at } 300 \text{ K}$		

**Marking**

①	
0	1
②	
0	1
③	
0	1
④	
0	1
⑤	
0	1

— End of the test —